DATA ITEM DESCRIPTION

Title: SYSTEM SAFETY PROGRAM PLAN (SSPP)

Number: DI-SAFT-82553

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Limitation: N/A

DTIC Applicable: No

GIDEP Applicable: No

Preparing Activity: 37 AF Safety Center

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Applicable Forms: None

Use/Relationship: A System Safety Program Plan (SSPP) documents the system safety methodology for the identification, classification, and mitigation of safety hazards as part of the overall Systems Engineering (SE) process. The SSPP should be an integral part of the Systems Engineering Management Plan (SEMP). The SSPP shall detail the tasks and activities that are required to implement a systematic approach of hazard analysis, risk assessment, and risk management. This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task required as delineated in the contract.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.
2. Format: The SSPP shall be in contractor's format, following the outline provided within this document.
3. Content: The Systems Safety Program Plan shall contain the following:

1. Scope and objectives. The contractor shall develop an SSPP to provide a basis of understanding between the Contractor and the Program Manager (PM) on how the safety hazard management effort will be integrated into the SE process. The SSPP shall describe, at a minimum: (1) the scope of the effort in terms of the system and its life-cycle, (2) the overall approach for accomplishing the General Requirements in Section 4 and other contractually required tasks, (3) integration of those efforts into SE and other Program Office management processes in order to support overall program objectives, and (4) resource requirements (funding, qualified personnel, and tools) to execute the SSPP. This Section shall account for all contractual hazard management requirements by providing a matrix that correlates these contractual requirements to the location(s) in the SSPP where each requirement is addressed.

2. SSPP interfaces. The SSPP shall:

   a. Identify the functional disciplines covered by the SSPP.

   b. Describe the SSPP interfaces between:

      (1) SE.

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(2) Other involved disciplines (e.g., logistics, maintainability, quality assurance, reliability, human factors engineering, transportability engineering, and medical support (health hazard assessments)).

3. Organization. The SSPP shall describe, at a minimum:

a. The organization or function of the system safety efforts within the SE process. Use charts to show organizational and functional relationships and lines of communication.

b. The staffing (manpower loading and schedule) of the system safety efforts by each of the involved functional disciplines and organizational units for the duration of the contract. The SSPP will identify responsibility and authority of each person and organizational unit involved in executing each of the contractual system safety requirements. The SSPP will also identify key personnel and provide a summary of the qualifications and credentials of the key system safety personnel. The SSPP will describe how and when the Contractor shall notify the Government prior to changes of key system safety personnel.

c. The procedures the contractor will use to integrate system-level and System-of-Systems (SoS) level hazard management efforts to the extent covered in the contract. These will include:

(1) Defining the roles of each associate contractor and subcontractor (and suppliers and vendors as applicable) to integrate safety requirements for the total system.
(2) Defining the safety interfaces between each associate contractor and subcontractor (and suppliers and vendors as applicable), e.g. integrating hazard analyses.
(3) Establishing Integrated Product Teams (IPTs) or Working Groups (WGs) with representatives from each associate contractor and subcontractor (and suppliers and vendors as applicable).
(4) Describing any specific SoS integration roles and responsibilities.
(5) Integrating hardware and software provided by the Government.
(6) Assigning requirements to action organizations and subcontractors.
(7) Coordinating subcontractor system safety engineering efforts.
(8) Facilitating safety reviews.
(9) Recommending mitigation measures; assessing feasibility, cost, and effectiveness of the measures; and allocating implementation responsibility to associate contractors and subcontractors.
(10) Reporting on program safety status and metrics.
(11) Describing procedures for addressing safety issues between associate contractors and subcontractors.

d. The process through which contractor management decisions will be made including timely notification of hazards with Catastrophic and Critical severity levels, as well as High and Serious risks to the Government; determining actions necessary in the event of mishaps, incidents, or malfunctions; and requesting waivers for safety requirements and program deviations.

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4. Milestones. The SSPP shall, at a minimum:

   a. Provide a schedule of system safety activities including required inputs and outputs and start and completion dates that support the SE process.

   b. Relate the system safety activities to integrated system-level activities (e.g., design analyses, tests, and demonstrations), technical reviews, program reviews, and major program milestones by recommending their inclusion in the Integrated Master Schedule (IMS).

   c. Identify the schedules for subsystem, component, and software activities applicable to the system safety activities but specified in other engineering studies and development efforts.

   d. Include a schedule of technical meetings between associate contractors and subcontractors to discuss, review, and integrate the safety effort.

5. General safety requirements and criteria. The SSPP shall:

   a. List the standards and system specifications containing safety requirements that the contractor shall use in the execution of the contract. Include titles, dates, and where applicable, paragraph numbers.

   b. Describe general engineering requirements and design criteria for safety risk management during system design and development.

   c. Identify safety risk management requirements, to include procedures, for test, operations and support, and disposal.

   e. Describe the method for ensuring flow-down of hazard identification and mitigation functions as well as associated requirements to subcontractors/suppliers.

6. Hazard analysis. At a minimum, the SSPP shall:

   a. Describe the processes for hazard identification, risk assessment, risk mitigation, risk communication, and support to risk acceptance.

      (1) For hazard identification, the SSPP shall describe the systematic identification process that evaluates the system throughout its life-cycle. This evaluation should include as a minimum system hardware and software, system interfaces (to include human interfaces), the intended use or application and operational environment, and disposal.
      (2) For risk assessment, the SSPP shall list the severity categories, probability levels, and Risk Assessment Codes (RAC) that shall be followed. The definitions located in MIL-STD-882E Section 4.3.3. (Copies of this document are available online at https://quicksearch.dla.mil.) assess

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(3) and document risk Tables I and II, and the RACs in Table III shall be used, unless tailored alternative definitions and/or a tailored matrix are formally approved in accordance with Department of Defense (DoD) Component policy.
(4) For risk mitigation, the SSPP shall describe how decisions will be made within the overall SE process. The SSPP shall prioritize the goal to eliminate hazards. When a hazard cannot be eliminated, the SSPP should describe the process for determining how the associated risk could be reduced to the lowest acceptable level within the constraints of cost, schedule, and performance by applying the system safety design order of precedence described in Section 4 of this Standard. SE process decisions on which mitigations to pursue will be the result of trade-off discussions between the involved technical disciplines.
(5) For risk acceptance, the SSPP shall describe the plan to address Government risk acceptance to include the procedures for communicating to the Government that a risk acceptance decision is required and providing the risk assessment documentation. In addition, the plan should include the procedures the Government has provided on how the Government will communicate to the Contractor the results of the proposed risk acceptance decision. In accordance with Department of Defense Instruction (DoDI) 5000.02, the Government may have to accept an event risk at multiple points in the life-cycle. (Copies of this document are available online at https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500002p.PDF)

b. Describe the approach for applying safety risk management to the use of Commercial- Off-the-Shelf (COTS), Government-Off-the-Shelf (GOTS), Non-Developmental Item (NDI), Government-Furnished Equipment (GFE), and Government-Furnished Information (GFI).

c. Describe closed-loop procedures for tracking and reporting identified hazards and associated risks, including those involving COTS, GOTS, NDI, GFE, and GFI. Include a detailed description of the Hazard Tracking System (HTS).

d. Describe the process for determining whether a qualitative or quantitative risk assessment is appropriate for a given hazard.

e. Identify the hazard analyses to be performed (e.g., Preliminary Hazard Analysis [PHA], Subsystem Hazard Analysis [SSHA]), analytical techniques to be used (e.g., Fault Tree Analysis [FTA], Failure Modes and Effects Criticality Analysis [FMECA]), and documentation of the results in the HTS.

f. Identify the scope of each analysis, integration of associate contractor and subcontractor hazard analyses with overall system hazard analyses, and the depth within the system that each analytical technique will be used.

g. When conducting SoS hazard analyses, the plan shall describe how analysis of the integrated system design, operations, and the interfaces between the products of each associate contractor or subcontractor and the end item will be executed. Data

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or analyses provided by associate contractors and subcontractors (and suppliers and vendors as applicable) shall be used in the conduct of this effort.

h. Describe the efforts to identify and control hazards associated with materials used-cycle.

i. Describe a systematic software system safety approach to:

(1) Identify and describe the software contributions to system hazards.
(2) Identify safety-significant (safety-critical and safety-related) software functions and software requirements.
(3) Identify the safety requirements associated with safety-significant software components and safety-related items.
(4) Identify and assign the Software Criticality Index (SwCI) for each safety-significant software function (SSSF) and its associated requirements.

7. Supporting data. At a minimum, the SSPP shall:

a. Describe the approach for collecting and processing pertinent hazard, mishap, and lessons learned data. This should include both historical data from similar or legacy systems used to assist in hazard identification and associated risk assessment, and current system data, e.g., the HTS.

b. Identify all documents or other media incorporating hazard management data by title, contract number, date(s) of delivery, and proposed means of delivery (hard copy, electronic, or real-time access) intended to be delivered to the Government under this contract, including documents or other media with other than unlimited rights for the Government. At a minimum, deliverable data shall include HTS data provided during contract execution and at contract closeout.

8. Verification and validation. At a minimum, the SSPP shall document how the safety risk management effort will:

a. Verify, validate, and document effectiveness of mitigation measures in reducing risk through test, analysis, inspection, etc.

b. Verify, validate, and document that hardware, software, and procedures comply with identified hazard management requirements.

c. Identify requirements for certifications, independent review board evaluations, and special testing (e.g., insensitive munitions tests and render-safe/emergency disposal procedures).

d. Ensure procedures are in place to transmit verification and validation information to the Government.

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9. Audit program. The SSPP shall describe the techniques and procedures to be employed by the contractor to make sure the requirements of the system safety process, as described in Section 4 of this Standard, are being accomplished.

10. Training. The SSPP shall describe the training for the personnel involved with the system safety process.

11. Incident reporting. The contractor shall describe in the SSPP the incident (especially mishap and malfunction) alerting, investigation, and reporting processes, including notification of the Government.

12. Details to be specified. The Request for Proposal (RFP) and Statement of Work (SOW) shall include the following, as applicable:

    a. Imposition of Task 102 in MIL-STD-882E.

    b. Identification of functional discipline(s) to be addressed by this task.

    c. Identification of any SoS requirements covered by this task, to include interfacing hardware and software provided by the Government.

    d. Requirements and methodology for submittal, review, and approval of this plan.

    e. Procedures for communicating formal Governmental risk acceptance to the contractor.

    f. Qualification requirements for key functional personnel.

    g. Other specific safety risk management requirements, e.g., specific risk definitions and matrix to be used on this program.

End of DI-SAFT-82553

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